INCH-POUND
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9 June 1997
SUPERSEDING
MIL-R-55182/13
4 October 1990

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, FILM, NONESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, AND SPACE LEVEL, STYLE RNC77

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for style RNC77, nonestablished reliability, established reliability, and space level, fixed, film, resistors. 1/

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-55182 - Resistors, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5049.)

1/ Third letter is variable, dependent upon lead material or capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications - Electronics Command and Fort Monmouth, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905
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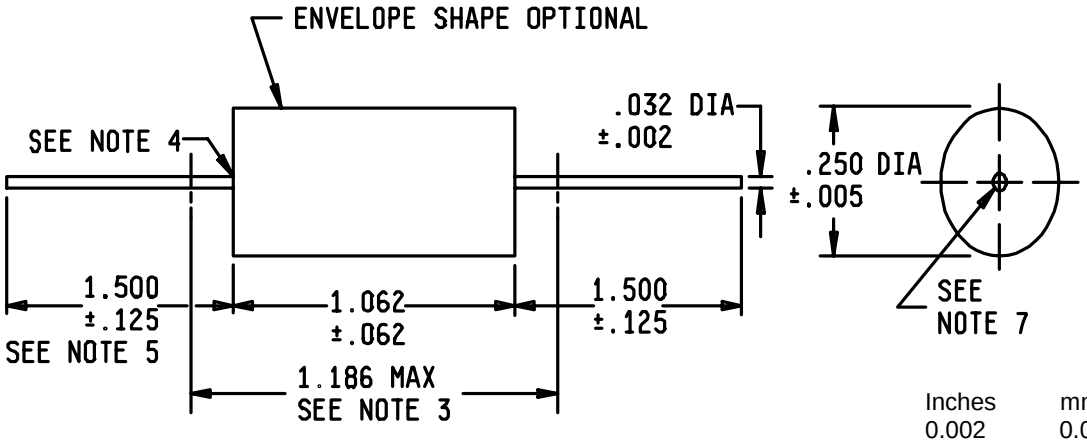

Inches	mm
0.002	0.05
0.005	0.13
0.016	0.41
0.031	0.79
0.032	0.81
0.062	1.57
0.125	3.18
0.250	6.35
0.625	15.88
1.062	26.97
1.186	30.12
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The maximum length is "clean lead" to "clean lead".
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.
5. Lead length for new design and tape and reel packaging shall be 1.00 inch +.625 inch, -.00 inch (25.4 mm +15.80 mm, -0.00 mm).
6. Lead concentric tolerance is to be measured at the point of lead egress from the resistor body.
7. Lead to be concentric to the resistor body within 0.016 TIR.

FIGURE 1. Style RNR77 resistors. 2/

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-55182G.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1.

3.2.1 Characteristic. Style RNR77 is available in characteristic C, characteristic E, characteristic H, characteristic J, and characteristic K.

3.2.2 Terminal. Terminal type RNC is available in characteristic H, characteristic J, and characteristic K only. Terminal type RNR is available in characteristic C and characteristic E only.

3.3 Power rating. The power rating shall be 1.0 watt at 125°C (see 6.3).

2/ Reference footnote 1/.

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3.4 Voltage rating. The continuous working voltage shall not exceed 750 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

Minimum resistance

Characteristic:

H, J, and K 3/ - - - - - - - - - -	1.0 ohms
C and E - - - - - - - - - - - - - -	20.0 ohms

Maximum resistance

Characteristic:

C and E - - - - - - - - - - - - - -	20.0 megohms
H, J, and K - - - - - - - - - - -	20.0 megohms

3.6 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±0.0005 percent per volt.

4. VERIFICATION

4.1 Verification. Verification shall be in accordance with MIL-PRF-55182.

4.2 Overload. The overload test duration shall be 1 hour with 2.25 x rated power overload; maximum voltage shall be 1,000 volts.

4.3 Terminal strength. The applied force shall be 5 pounds.

4.4 Dielectric withstanding voltage. The magnitude of test voltage shall be 900 volts for atmospheric pressure and 450 volts for barometric pressure.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-55182 are applicable to this specification.

3/ For resistance tolerance D and resistance tolerance F only. For resistance tolerance B, the minimum resistor value shall be 10.00 ohms.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

6.3 Power rating at 70°C. The power rating at 70°C for 2,000 hours life test has been established at 2.0 watts; however, it should be noted that the failure rate level is established at the single condition of 125°C, 1.0 watt and 10,000 hours life test duration. The user should consider the FR level as assigned only to the 125°C.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5905-1458-09)

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